

Reversing Protectionism: A First Look at Product-level Trade Data from Smoot-Hawley to the GATT*

Supplementary Online Appendix

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April 24, 2026

In this supplementary appendix, we expand on the information about the data set contained in the Data Appendix of “Reversing Protectionism: A First Look at Product-level Trade Data from Smoot-Hawley to the GATT (Section 8).” We start by presenting an example of a product across different classification systems in different years. This provides context about our data sources and the information we obtain from each source. We further discuss the digitization and merging processes we used to create our final data set. Finally, we discuss some products whose tariffs are created by formula instead of being individually negotiated and how we deal with them in our data.

*The views expressed in this paper are strictly those of the authors and do not represent the opinions of the U.S. International Trade Commission or any of its Commissioners.

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1 An Example: *Rye* from 1930 to 2024

In order to more accurately describe our data sources and demonstrate the evolution of the product classification code system across systems and over time, we pick a representative product *Rye*, and present how it is described in various historical documents. Smoot-Hawley marks the beginning of our sample period. At that time *Rye* was included in Paragraph number 728 as shown in Figure 1. The rate for *Rye* was 15 cents per bushel of fifty-six pounds. The same product was assigned Schedule A code 190.23 in 1930 in Figure 2.

An examination of the 1946 Schedule A, as illustrated in Figure 3, reveals a reduction in the tariff applicable to *Rye* from 15 cents to 12 cents per bushel. In addition to the tariff reduction, this figure shows three other features. First, the Schedule A code underwent a change from 109.23 to 1044000. This demonstrates one of the primary challenges in constructing panel data in this context: that many product codes change over time. We discuss this in Section 8.2.2. Second, the original Smoot-Hawley rate is documented in column 1, which is analogous to the “Column 2” in the TSUSA system and the Harmonized System (HS). It is, however, important to point out that this rate does not always equal the original Smoot-Hawley rate due to the presence of the non-trade-agreement provisions discussed Section 8.3.4, such as Section 336. Finally, the designation “Can.” in the second column of the table is an abbreviation for Canada. It indicates that this tariff reduction is the result of the bilateral trade agreement between the U.S. and Canada. The negotiation of bilateral trade agreements constituted a significant instrument employed by the U.S. to reduce tariff rates prior to the establishment of the GATT, and by digitizing these documents, our data set provides a record of products involved in bilateral agreements and their tariff

reduction.

For the GATT tariff commitments, we use the detailed tariff schedules from the United Nations Treaty Collection (UNTC) series to supplement our other sources.¹ Figure 4 illustrates the impact of Geneva round negotiations on *Rye*, which is covered under Paragraph 728 in the Smoot-Hawley system. During the course of the negotiations, the tariff rate applicable to *Rye* was further reduced to 6 cents per bushel of fifty-six pounds, a decrease from the 12 cents per bushel of fifty-six pounds previously agreed with Canada in the bilateral agreement.

Finally, we show how the TSUSA and HS systems account for *Rye* to demonstrate how our data could be linked to more modern product classification systems. Figures 5 and 6 show that *Rye* was assigned code 130.60 in TSUSA 1963 with no change in the tariff and is assigned code 1002.90 and becomes duty-free under the latest version of the Harmonized Tariff Schedule. Notice that in both pictures, the “Column 2” rates are exactly the original Smoot-Hawley rate of 15 cents per bushel of fifty-six pounds.²

Rice.	PAR. 727. Paddy or rough rice, $1\frac{1}{4}$ cents per pound; brown rice (hulls removed, all or in part), $1\frac{1}{2}$ cents per pound; milled rice (bran removed, all or in part), $2\frac{1}{2}$ cents per pound; broken rice, which will pass readily through a metal sieve perforated with round holes five and one-half sixty-fourths of one inch in diameter, and rice meal, flour, polish, and bran, five-eighths of 1 cent per pound.
Rye.	PAR. 728. Rye, 15 cents per bushel of fifty-six pounds; rye malt, 40 cents per one hundred pounds; rye flour and meal, 45 cents per one hundred pounds.
Wheat.	PAR. 729. Wheat, 42 cents per bushel of sixty pounds; wheat, unfit for human consumption, 10 per centum ad valorem; wheat flour, semolina, crushed or cracked wheat, and similar wheat

Figure 1: Rye in 1930 Smoot-Hawley Act

¹See Table 12 in Section 8.1.1 for details.

²The HTS changes the unit of measure for *Rye* from bushels of fifty six pounds to kilograms. To see that the “Column 2” rate is the same here as the Smoot-Hawley rate, note that 56 pounds is equal to approximately 25.401 kilograms.

Group 1.—VEGETABLE FOOD PRODUCTS AND BEVERAGES

Class No.	Commodity	Unit of quantity	Rate of duty	Tariff paragraph
GRAINS AND PREPARATIONS				
10310	Corn.....	Bu. ¹	25¢ bu.....	724
10410	Oats.....	Bu. ²	16¢ bu.....	726
	Rice:			
10510	Paddy.....	Lb.....	1¼¢ lb.....	727
10511	Uncleaned or brown rice.....	Lb.....	1½¢ lb.....	
10530	Cleaned or milled rice.....	Lb.....	2¼¢ lb.....	
10540	Patna rice, cleaned, for use in canned soups.....	Lb.....	Free.....	1752
10590	Broken rice and rice meal, flour, polish and bran.....	Lb.....	⅝¢ lb.....	727
	Wheat:			
10700	For grinding in bond and export.....	Bu. ³	Free.....	Sec. 311
10710	Ground into flour for export to Cuba.....	Bu. ³	State duty.....	
10711	Other n. s. p. f.....	Bu. ³	42¢ bu.....	
10720	Wheat flour.....	Lb.....	1.04¢ lb.....	729
10750	Biscuits, wafers, puddings, cakes, and other baked articles.		30%.....	733
	Macaroni, vermicelli, noodles, and similar pastes:			
10770	Containing no eggs or egg products.....	Lb.....	2¢ lb.....	725
10771	Containing eggs or egg products.....	Lb.....	3¢ lb.....	
	Other grains and preparations:			
10790	Bread, yeast-leavened.....	Lb.....	Free.....	1623
109. 11	Buckwheat.....	Lb.....	0.25¢ lb.....	723
109. 12	Buckwheat flour and grits.....	Lb.....	½¢ lb.....	
	Barley—			
109. 13	Hulled or unhulled.....	Bu. ⁴	20¢ bu.....	722
109. 14	Flour and patent.....	Lb.....	2¢ lb.....	
109. 15	Pearl.....	Lb.....	2¢ lb.....	
109. 16	Malt.....	Lb.....	0.4¢ lb.....	
109. 17	Cereal breakfast foods and preparations, n. s. p. f.....	Lb.....	20%.....	732
109. 18	Cracked corn.....	Bu. ¹	25¢ bu.....	724
109. 19	Corn meal, flour, grits, and similar products.....	Lb.....	0.5¢ lb.....	
109. 20	Semolina, crushed, cracked, and other wheat products, n. s. p. f.....	Lb.....	1.04¢ lb.....	729
109. 21	Oats, unhulled, ground.....	Lb.....	0.45¢ lb.....	726
109. 22	Oatmeal, rolled, grits, and similar oat products.....	Lb.....	0.8¢ lb.....	
109. 23	Rye.....	Bu. ¹	15¢ bu.....	728
109. 25	Rye malt.....	Lb.....	0.4¢ lb.....	
109. 26	Rye flour and meal.....	Lb.....	0.45¢ lb.....	
109. 27	Grains, n. s. p. f. (<i>specify by name</i>).....	Lb.....	10%.....	1558
109. 28	Grain flour, n. s. p. f. (<i>specify by name</i>).....	Lb.....	20%.....	
109. 30	Wheat unfit for human consumption.....	Bu. ³	10%.....	729

Figure 2: Rye in 1930 Schedule A

2 Digitization Process

In this section, we provide details about how we digitized and merged our data sources.

Group 1.—VEGETABLE FOOD PRODUCTS AND BEVERAGES

Commodity number	Commodity description and economic class	Unit of quantity	Rate of duty		Tariff paragraph
			1930 Tariff Act (except as noted)	Trade agreement	
GRAINS AND PREPARATIONS					
1020 000	Barley (bushel 48 pounds).....	(2) Bu.....5	20¢ bu.....	15¢ bu. Can.....	722
1025 000	Buckwheat.....	(2) Lb.....1	0.25¢ lb.....	0.15¢ lb. Can.....	723
1031 000	Corn (bushel 56 pounds).....	(2) Bu.....5	25¢ bu.....	10¢ bu. Cuba.....	724
1041 000 ^a	Oats, hulled or unhulled, whether or not fit for human consumption (bushel 32 pounds).	(2) Bu.....5	16¢ bu.....	8¢ bu. Can.....	726
1044 000	Rye (bushel 56 pounds).....	(2) Bu.....5	15¢ bu.....	12¢ bu. Can.....	728
Rice:					
1051 000	Paddy.....	(2) Lb.....1	1 3/4¢ lb.....		727
1051 100	Uncleaned or brown rice.....	(2) Lb.....1	1 3/4¢ lb.....		
1053 000	Cleaned or milled rice.....	(4) Lb.....1	2 3/4¢ lb.....		
1054 000	Patna rice, cleaned, for use in canned soups.	(4) Lb.....1	Free.....		1752
1059 100	Rice meal, flour, polish, and bran.....	(4) Lb.....1	5/8¢ lb.....		727
1059 200	Broken rice (Include brewers' rice).....	(4) Lb.....1	5/8¢ lb.....	5/16¢ lb. Neth.....	
1060 000 ^a	Wheat (bushel 60 pounds): For grinding in bond and export to countries other than Cuba.	(2) Bu.....5	Free.....		Sec. 311
1065 000 ^a	Ground into flour for export to Cuba.....	(2) Bu.....5	Report duty collected quar-		

Figure 3: Rye in 1946 Schedule A

2.1 The Smoot-Hawley Paragraph Number System

We collect the U.S. negotiated tariff data for the GATT based on the Smoot-Hawley paragraph number system from the original copy of the Smoot-Hawley Act in 1930 and the United Nations Treaty Collection (UNTC), which contains the consolidated tariff schedule for each participating country and each GATT round. We initially collated and digitized the tariff schedules for the Geneva, Annecy, and Torquay rounds for the United States, but in this paper we are only using the Geneva tariff schedule. We referenced the Smoot-Hawley Act whenever necessary to concord between 1930 and 1947 and to infer which products were subject to non-trade-agreement provisions.

The U.S. bilateral trade agreements with other countries also played an important role in the liberalization of trade prior to the GATT. The [Office of the Historian](#) website maintained by the U.S. Department of State collects all the related information of negotiations/correspondence between the two nations for each agreement. We sourced the texts of the bilateral trade agreements from the Federal Register as compiled in the [United States](#)

[Statutes at Large](#). These texts include full product detail on the tariffs negotiated in each bilateral agreement.

The first document we found is a tariff schedule for the U.S. that consolidates the concessions made in the Geneva, Annecy and Torquay rounds. We refer to this as the “consolidated Torquay schedule.” This document was in hard copy and borrowed from the University of Texas Libraries. We scanned the hard copy, conducted optical character recognition (OCR) and digitized the file in R. By running the R package *pdftools*,³ we obtain an editable Microsoft Excel file that consists of detailed product descriptions and their corresponding tariff rates. Since the consolidated Torquay schedule includes the tariff commitments of the first three rounds of the GATT, we were able to construct a benchmark schedule that includes most of the products from Smoot-Hawley to the Torquay round (1950).

Subsequently, we found more complete and systematic data in the UNTC, which provides the individual schedule of products negotiated in each round of the GATT in the format of Smoot-Hawley paragraph number. We then manually entered the U.S. tariff rate for each round line by line based on the framework we had constructed via the consolidated schedules. To check the reliability of our benchmark file, we compared the three individual rounds we have from the UNTC with the consolidated version by the Torquay round and found no discrepancies in the reported tariff rates.

Next, we located a scanned copy of the 1930 Smoot-Hawley Tariff Act and integrated its tariff rates into our dataset. The Smoot-Hawley tariffs make sense as a benchmark because they continued to be the prevailing legal tariffs of the U.S. unless modified by subsequent

³<https://cran.r-project.org/web/packages/pdftools/pdftools.pdf>

agreement or legislation. We therefore made sure to enter all products in the original Smoot-Hawley tariffs into our data. That is, even if some products did not receive commitments in later rounds, they are still included in our dataset.

2.2 Schedule A System

For more accurate analysis, our final dataset is organized according to the Schedule A system. The Schedule A system has more precise descriptions and divides products into smaller categories than the Smoot-Hawley paragraph number system. It was used to process imports and generate the import data during this time period, so organizing by Schedule A allows us to construct our final data set at a detailed product level while integrating tariff rates with import quantities and values. Information pertaining to the Schedule A system is principally derived from the U.S. Department of Commerce annual report, entitled “Schedule A, Statistical Classification of Imports into the United States.” The document was published for nonconsecutive years (e.g., in 1933 but not in 1934 or 1935, and then again in 1936 and 1937). Although the HathiTrust digital library contains a substantial number of Schedule A reports from 1930 to 1946, its collection is not comprehensive. Consequently, we also utilized the interlibrary loan system through the Syracuse University library to obtain and digitize the remaining reports. We verify and extend these sources with the import records from the Department of Commerce and the texts of the bilateral trade agreements.

We digitized the Schedule A series by hand using double data entry procedures to ensure accuracy. The 1946 Schedule A document contains all tariff changes between 1930 and 1946 at product level in terms of the Schedule A codes, whether through bilateral negotiations

or unilateral action. We later also digitized other years of the Schedule A and created concordances to construct our panel data. Finally, by assigning a unique identifier to each product, we constructed a sorting system that allows the data to be sorted according to either the Smoot-Hawley paragraph number or the Schedule A code.

In Schedule A, the industrial classification of products are classified as eleven groups, which we present in Table 1. The groups are further divided into subgroups that define the products classification at a more disaggregated level. For example, in “Group 4: Wood and Paper,” there are six subgroups: “Wood, unmanufactured,” “Sawmill products (lumber),” “Wood manufactures,” “Cork and manufactures,” “Paper base stocks,” and “Paper and manufactures.” Altogether, there are 93 subgroups under the 11 groups. We use the 93 subgroup definitions in the 1950 Schedule A—after verifying the group definitions are the same in the 1946 Schedule A—as industry-level fixed effects for the terms-of-trade analysis.

Table 1: Schedule A Codes: Groups

Schedule	Category	Schedule A Code
00	Animals and Animal Products, Edible	0010600-0097500
0	Animals and Animal Products, Inedible	0201000-0990290
1	Vegetable Food Products and Beverages	1020000-1900000
2	Vegetable Products, Inedible, Except Fibers and Wood	2011000-2960150
3	Textile Fibers and Manufactures	3001000-3981100
4	Wood and Paper	4007100-4799990
5	Nonmetallic Minerals	5000000-5958100
6	Metal and Manufactures Except Machinery and Vehicles	6001000-6900500
7	Machinery and Vehicles	7063000-7940290
8	Chemicals and Related Products	8000000-8731200
9	Miscellaneous	9001000-9990750

2.3 Industrial Classification Systems

In this section, we compare the industrial classification across different systems. The industrial groupings were defined differently under Smoot-Hawley (“Schedules”) and the Schedule A (“Groups”). We provide the comparison between these two systems in Tables 2 and 1. We also include an example of the linkage across schedules (groups) between the Smoot-Hawley Paragraph Number and the Schedule A Code in Figure 7. To provide further comparison of such systems with the modern system, we include a summary of the TSUSA “Sections” in Table 3 as demonstration.

Table 2: Schedules in Smoot-Hawley

Schedule	Category	Paragraph Number
1	Chemicals, Oils, and Paints	1 to 97
2	Earths, Earthenware, and Glassware	201 to 236
3	Metals and Manufactures of	301 to 398
4	Wood and Manufactures of	401 to 412
5	Sugar, Molasses, and Manufactures of	501 to 506
6	Tobacco and Manufactures of	601 to 605
7	Agricultural Products and Provisions	701 to 783
8	Spirits, Wines, and Other Beverages	801 to 815
9	Cotton Manufactures	901 to 924
10	Flax, Hemp, Jute, and Manufactures of	1001 to 1022
11	Wool and Manufactures of	1101 to 1122
12	Silk Manufactures	1201 to 1211
13	Manufactures of Rayon or Other Synthetic Textile	1301 to 1313
14	Papers and Books	1401 to 1413
15	Sundries	1501 to 1559
16,17,18	Title II - Free List	1601 to 1814

- 79 -

SCHEDULE XI — UNITED STATES OF AMERICAPart I (continued)

Tariff Act of 1930, para- graph	Description of Products	Rate of Duty
722	Barley, hulled or unhulled.....	7-1/2¢ per bu. of 48 lb.
722	Barley malt.....	30¢ per 100 lb.
722	Pearl barley.....	1/2¢ per lb.
722	Patent barley and barley flour.....	2¢ per lb.
723	Buckwheat, hulled or unhulled.....	10¢ per 100 lb.
723	Buckwheat flour and grits or groats.....	1/5¢ per lb.
726	Oats, hulled or unhulled.....	4¢ per bu. of 32 lb.
726	Unhulled ground oats.....	25¢ per 100 lb.
726	Oatmeal, rolled oats, oat grits, and similar oat products.....	10% ad val., but not less than 40¢ nor more than 80¢ per 100 lb.
727	Broken rice, which will pass readily through a metal sieve perforated with round holes five and one-half sixty-fourths of one inch in diameter.....	5/16¢ per lb.
727	Rice meal, flour, polish, and bran.....	5/16¢ per lb.
728	Rye.....	6¢ per bu. of 56 lb.
728	Rye malt.....	30¢ per 100 lb.
728	Rye flour and meal.....	30¢ per 100 lb.
729	Wheat.....	21¢ per bu. of 60 lb.
729	Wheat, unfit for human consumption.....	5% ad val.

Figure 4: Rye in Geneva Round U.S. Tariff Schedule from UNTC

TARIFF SCHEDULES OF THE UNITED STATES ANNOTATED (1963)

SCHEDULE 1. - ANIMAL AND VEGETABLE PRODUCTS
Part 7. - Cereal Grains, Milled Grain Products, and Malts and Starches

34

Item	Stat. Sur- fix	Articles	Units of Quantity	Rates of Duty	
				1	2
130.45	00	Oats, hulled or not hulled.....	Bu..	4¢ per bu. of 32 lbs.	16¢ per bu. of 32 lbs.
130.50	00	Rice:	Lb.....	1.25¢ per lb.	1.25¢ per lb.
130.55	00	Paddy or rough.....	Lb.....	1.5¢ per lb.	1.5¢ per lb.
130.60	00	Brown (hulls removed, all or in part).....	Bu.....	6¢ per bu. of 56 lbs.	15¢ per bu. of 56 lbs.
130.65	20	Wheat:		5% ad val.	10% ad val.
	40	Not fit for human consumption.....	Bu.....		
130.70	20	Seed wheat.....	Bu.....	21¢ per bu. of 60 lbs.	42¢ per bu. of 60 lbs.
	40	Other.....	Bu.....		
Subpart B. - Milled Grain Products ^{1/}					
Subpart B headnote:					
1. The term "milled grain products", as used in this subpart, embraces flours, grits, groats, meal, flaked or rolled grains, and other products, all the foregoing, whether or not fit for human consumption, made or derived from the grains named in subpart A of this part by grinding, crushing, breaking, rolling, flaking, pearling, polishing, or similar milling processes, but does not include by-products or wastes resulting from any of these processes.					
Milled grain products:					
Fit for human consumption:					
131.10	00	Barley:	Lb.....	0.4¢ per lb.	2¢ per lb.
131.12	00	Pearl barley.....	Lb.....	2¢ per lb.	2¢ per lb.
131.15	00	Other.....	Lb.....	0.2¢ per lb.	0.5¢ per lb.
131.20	00	Buckwheat.....	Cwt.....	50¢ per 100 lbs.	50¢ per 100 lbs.
131.21	00	Corn.....		40¢ per 100 lbs. (s)	
131.25	00	If product of Cuba.....	Cwt.....	10% ad val.	80¢ per 100 lbs.
131.27	00	Oats:	Cwt.....	80¢ per 100 lbs.	80¢ per 100 lbs.
131.30	00	Valued not over \$8 per 100 lbs.....	Lb.....	2.5¢ per lb.	2.5¢ per lb.
131.31	00	Valued over \$8 per 100 lbs.....		2¢ per lb. (s)	
131.33	00	Rice:	Lb.....	0.3125¢ per lb.	0.625¢ per lb.
		Milled rice (bran removed all or in part).....	Lb.....	0.3125¢ per lb.	0.625¢ per lb.
		If product of Cuba.....	Lb.....	Free	Free
		Broken rice, which will pass readily through a metal sieve perforated with round holes 11/128 inch in diameter.....	Cwt.....	22.5¢ per 100 lbs.	45¢ per 100 lbs.
131.35	00	Meal and flour.....	Cwt.....	52¢ per 100 lbs.	\$1.04 per 100 lbs.
131.37	00	Patna, cleaned, for use in the manufacture of canned soups.....	Lb.....	20% ad val.	20% ad val.
131.38	00	Rye.....		16% ad val. (s)	
131.40	00	Wheat.....	Cwt.....	15¢ per 100 lbs.	40¢ per 100 lbs.
131.45	00	Other.....	Cwt.....	10¢ per 100 lbs.	25¢ per 100 lbs.
131.46	00	If products of Cuba.....	Cwt.....	45¢ per 100 lbs.	45¢ per 100 lbs.
131.50	00	Not fit for human consumption:		18¢ per 100 lbs. (s)	
131.57	00	Barley.....	Cwt.....	12.5¢ per 100 lbs.	50¢ per 100 lbs.
131.60	00	Buckwheat.....	Cwt.....	31.25¢ per 100 lbs.	62.5¢ per 100 lbs.
131.61	00	Corn.....	Cwt.....	10.5¢ per 100 lbs.	26.8¢ per 100 lbs.
131.65	00	If product of Cuba.....			
131.67	00	Oats.....			
131.69	00	Rice.....			
131.70	00	Rye.....			
131.72	00	Wheat:			
131.75	00	Flour.....	Cwt.....	2.5% ad val.	10% ad val.
131.80	00	Other.....	Cwt.....	5% ad val.	10% ad val.
		Other.....	Cwt.....	10% ad val.	20% ad val.

(s) = Suspended. See general headnote 3(b).
^{1/} Imports of certain milled grain products are subject to additional import restrictions. See Appendix to Tariff Schedules.

Figure 5: Rye in TSUSA 1963

Harmonized Tariff Schedule of the United States Revision 8 (2024)

Annotated for Statistical Reporting Purposes

II
10-3

Heading/ Subheading	Stat. Suf- fix	Article Description	Unit of Quantity	Rates of Duty		
				1		2
				General	Special	
1001 (con.)		Wheat and meslin: (con.)				
1001.91.00	00	Other:				
		Seed.....	kg.....	2.8% ^{2/}	Free (A+, AU, BH, CL, CO, D, E, IL, JO, KR, MA, OM, P, PA, PE, S, SG)	10%
1001.99.00		Other.....		0.35¢/kg ^{4/}	Free (A+, AU, BH, CL, CO, D, E, IL, JO, KR, MA, OM, P, PA, PE, S, SG)	1.5¢/kg
	05	Canadian western extra strong hard red spring (CWES/HRS) wheat.....	kg			
		Other:				
	07	Certified organic:				
		Red spring wheat.....	kg			
	09	Other.....	kg			
		Other:				
		Red spring wheat:				
		Grade 1:				
	10	Having a specified protein content not exceeding 12.9% by weight....	kg			
	17	Having a specified protein content exceeding 12.9% but not exceeding 13.9% by weight.....	kg			
	18	Having a specified protein content exceeding 13.9% by weight.....	kg			
		Grade 2:				
	27	Having a specified protein content not exceeding 12.9% by weight....	kg			
	30	Having a specified protein content exceeding 12.9% but not exceeding 13.9% by weight.....	kg			
	33	Having a specified protein content exceeding 13.9% by weight.....	kg			
	36	Other red spring wheat.....	kg			
	45	White winter wheat.....	kg			
	55	"Canadian" western red winter wheat.....	kg			
	65	Soft white spring wheat.....	kg			
	97	Other.....	kg			
1002		Rye:				
1002.10.00		Seed.....		Free ^{2/}		0.59¢/kg
	10	Seeds of a kind used for sowing.....	kg			
	90	Other.....	kg			
1002.90.00	00	Other.....	kg.....	Free ^{2/}		0.59¢/kg

Figure 6: Rye in 2024 Harmonized System

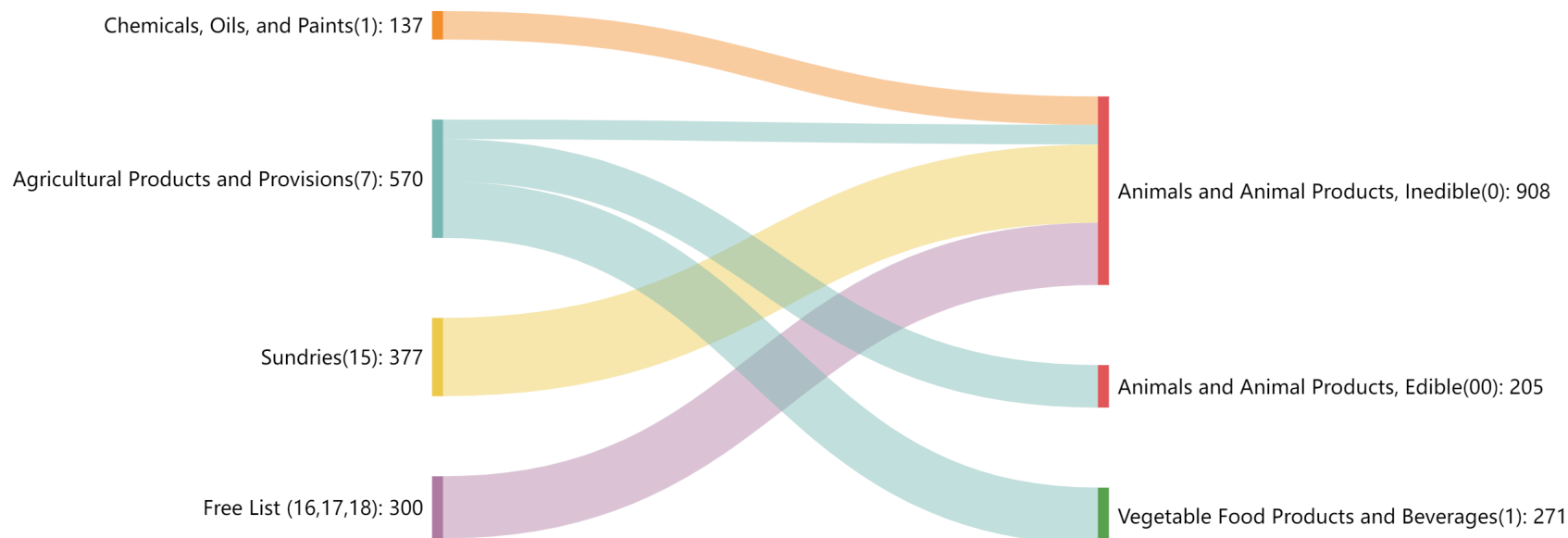


Figure 7: Relationship between Smoot-Hawley and Schedule A for Food products

Notes: The left-hand side of the chart shows where products are classified under the Smoot-Hawley system, while the right-hand side shows where they are classified under the Schedule A. The number after each colon is the number of products. For example, the first line on the left-hand side implies that there are 137 lines in Schedule 1 in Smoot-Hawley (Chemicals, Oils and Paints) that are concorded to Group 0 in the Schedule A system (Animals and Animal Products, Inedible).

Table 3: Sections in TSUSA

Section	Category	TSUSA.Code
1	Animal and Vegetable Products	100.01 to 193.25
2	Wood and Paper; Printed Matter	200.03 to 274.90
3	Textile Fibers and Textile Products	300.10 to 390.60
4	Chemicals and Related Products	401.02 to 495.20
5	Nonmetallic Minerals and Products	511.11 to 548.05
6	Metals and Metal Products	601.03 to 696.60
7	Specified Products: Miscellaneous and Nonemunerated Products	700.05 to 799.00
8	Special Classification Provisions	800 to 870.25

Table 4: Sections in HS

Section	Category	HS Code
I	Live animals; animal products	0101.21 to 0511.99
II	Vegetable products	0601.10 to 1404.90
III	Fats and oils	1501.10 to 1522.00
IV	Prepared foodstuffs; beverages; tobacco	1601.00 to 2403.99
V	Mineral products	2501.00 to 2716.00
VI	Chemical or allied industries	2801.10 to 3826.00
VII	Plastics; rubber	3901.10 to 4017.00
VIII	Leather; travel goods; animal gut	4101.20 to 4304.00
IX	Wood; cork; basketware	4401.11 to 4602.90
X	Pulp; paper and paperboard	4701.00 to 4911.99
XI	Textiles and textile articles	5001.00 to 6310.90
XII	Footwear; headgear; feathers; hair	6401.10 to 6704.90
XIII	Stone; ceramic; glass	6801.00 to 7020.00
XIV	Pearls; precious metals; jewellery; coin	7101.10 to 7118.90
XV	Base metals	7201.10 to 8306.99
XVI	Machinery; electrical equipment	8401.10 to 8548.00
XVII	Vehicles; aircraft; vessels	8601.10 to 8908.00
XVIII	Optical; medical; clocks; musical instruments	9001.10 to 9209.99
XIX	Arms and ammunition	9301.00 to 9307.00
XX	Miscellaneous manufactured articles	9401.10 to 9619.00
XXI	Works of art; antiques	9701.10 to 9706.00

2.4 Concordance

2.4.1 Concordance Between Schedule A and Smoot-Hawley

After digitizing the above documents, we concord the resulting data sets to create our data set. The initial step is to create a concordance between the Schedule A Codes and Smoot-Hawley Paragraph Numbers. This is achieved by manually entering the paragraph numbers for each Schedule A code from the 1946 Schedule A. A considerable number of Schedule A products often share the same Paragraph Number; in contrast, Schedule A codes virtually always maintain a one-to-one correspondence with the import records. To facilitate the sorting of products within a Smoot-Hawley paragraph, we assign a unique item number to each Schedule A tariff (for mixed tariffs, we enter one observation in our raw data for each element of the tariff).

A few Schedule A codes are associated with multiple paragraph numbers. In most such instances, the multiple paragraph numbers encompass a principal product description and relevant supplementary provisions, such as additional duties on a particular material the product is made of. We assign the paragraph number in these instances by manually comparing the text of the Smoot-Hawley paragraphs to find the paragraph number that is most important in classifying the product. This means that we can sort products based on the degree of similarity in their descriptions. In a few instances where the descriptions are too ambiguous to determine the primary paragraph number, we use the first paragraph number documented in the original government schedules. This item number system enables the sorting of the data and the potential merging with other data sets that employ the Schedule

A or Smoot-Hawley systems.

2.4.2 Concoring Schedule A Over Time

This subsection describes the methodology employed in the Schedule A concordance table. The concordance tables are constructed separately for the main analysis comparing tariff changes from 1933 to 1947, and for the elasticity estimation covering the period of 1935 to 1938. The methodology and steps are basically the same, and for simplicity, we will explain our steps on concoring for the main data from 1933 to 1947.

The initial step in concoring is to select the benchmark year, as the concordance can be constructed in two directions: from the latter year to the former year or from the former year to the latter year. The selection of a benchmark year faces a trade-off that later schedules tend to have more lines and a more organized structure, since new codes with more specific classifications are added over time as tariff negotiations insert more nuance into the U.S. tariff schedules. However, these new lines were not negotiated separately in previous agreements and hence did not have separate import records in earlier years.

Since the primary objective of this study is to examine the formation and impact of the inaugural round of GATT (Geneva 1947), we decided to use 1946, just before the Geneva round, as our benchmark for the construction of the concordance. Using 1946 as the benchmark year provides the most comprehensive pre-Geneva rates schedule, which encompasses all bilateral agreements up to the first GATT round. It also allows us to cleanly use the 1947 import data to calculate unit values for the Geneva round, as 1947 imports are enumerated using the 1946 Schedule A codes. This is true for exactly the same reason that we selected

1947 as the unit value year: they were collected before the Geneva round tariffs took effect. This means that the Schedule A code changes that were necessitated by the Geneva tariff commitments had not yet taken effect in 1947.

A prerequisite for constructing these concordances is ensuring that all Schedule A codes adhere to a uniform format. Since our Schedule A code identifiers use the more standardized 7-digit codes of 1946, we developed an algorithm to extend the 5-digit Schedule A code in 1933 to the 7-digit codes in 1946 and 1948. This mostly involves adding zeros and moving the location of the decimal point.

The Schedule A product list has undergone numerous substantive modifications over time, including alterations to its syntax and provisions, which creates a significant challenge in creating a comprehensive panel data set. In addition to the modifications to the Schedule A codes, another challenge is the mapping relationship between products over time. In practice, cases are classified as 1-to-1, M-to-1 (one or multiple old codes merge into one new code), 1-to-N (one old code splits into one or multiple new codes), or M-to-N (one or multiple old codes are mapped to one or multiple new codes). Some new products are added to the Schedule A, while some older products are also removed.

After setting the benchmark year, we consider five elements when connecting product codes across schedules in different years: the Schedule A number, unit of quantity, tariff rate, Smoot-Hawley paragraph number, and product description. We classify the matches into seven types, and describe them in terms of the 1933 to 1946 concordance.

Default: a product's schedule A code is unchanged and the description is substantively the same through the period.

Split: a product in the 1933 Schedule A was split into multiple products in the 1946 Schedule A.

Merge: multiple products in the 1933 Schedule A were merged into a single product in the 1946 Schedule A.

New: a new product that does not appear in the 1933 Schedule A was created.

Obsolete: a product in the 1933 Schedule A was not provided for in the 1946 Schedule A.

Reassign: a product with the same substantive description as in the 1933 Schedule A was assigned a new code in the 1946 Schedule A.

Mixed: an M-N match that does not fit one of the other types. This type of mapping is not frequent in the sample, and is usually accompanied by a change in the logic of the chapter and rephrasing of the descriptions.

Assigning a concordance type to each match offers several benefits: it provides a comprehensive view of the expanding range of products in the Schedule A and is essential for linking the Schedule A tariff data with import data to calculate AVEs. We made these assignments for 1933 to 1946 and for 1946 to 1948.

3 Formulaic Tariff Rates

In several cases, one negotiated tariff change affects many products via a formula. We keep only representative products in these cases so as not to over-state the scope of the negotiated change in tariffs. This means that we include only representative lines for two

types of products: Cotton textiles—and the handkerchiefs made from them—and the alloy content of some iron and steel products.

3.1 Alloy Content of Iron and Steel Products

Some metal products exhibit a high degree of diversity in their alloy content. The structure of the tariff rates for these metal products is comprised of three distinct rates. The first is the base rate, which applies to products manufactured from steel. The second is an additional rate, which is applied to products manufactured from steel that have been specially processed. This is indicated by the digit “2” in the sixth position of the code. The third is a separate rate, which is applied to products containing an alloy. This is indicated by the seventh digit in the code, and charged only on the net content. To take one example, Schedule A code 6008820 is defined as “*steel bars, whether solid or hollow, valued over 16 cents per pound and if galvanized or coated with metal.*” Its rate is comprised of a 20 percent ad valorem (base rate) plus a specific rate of 2 cents per pound. A related line, 6008824, indicates the *chromium content* contained in this steel bar product has a tariff rate of 3 cents per pound on the weight of the alloy content. We exclude almost all of these alloy content lines (approximately 100 Schedule A numbers), keeping only one representative Schedule A number for each type of alloy.⁴

⁴Note that most of these lines would not be included in our data set even without this restriction because most of them have specific tariff rates but only quantities are included in the import data so that we cannot calculate their AVEs.

3.2 Cotton Textiles

The tariffs for cotton yarn and cotton cloth textile products are contingent on the process by which they are made, the presence of other materials, and size (yarn number). The structure of textile products, as defined by Schedule A codes from 3010 to 3067 (excluding 3030 000 and 3030 100), takes account of these components systematically. The initial element is a base rate related to the type of yarn or cloth (i.e., bleached, unbleached, dyed). This is denoted by a fourth digit of zero in the Schedule A number. The second component is the cumulative rate, which is contingent upon the yarn number of the product and is reflected in the final three digits of the Schedule A code. The third component denotes the amount of non-cotton content and the method of production and is reflected in the fourth digit of the product. To illustrate, consider product 3061005. The base rate is 3060, which has a 16 percent ad valorem tariff. The yarn number is five, which adds up to $16 + 0.35 \times 5 = 17.75$ percent ad valorem rate. Additionally, the fourth digit is equal to one, categorizing the product as “Woven with 8 more harness, or with Jacquard, lappet, or swivel attachments,” which adds an additional 10 percent rate. Thus, the ultimate rate for product 3061005 in the context of the Smoot-Hawley tariff is 27.75 percent.

The challenge of dealing with textile products is that there are approximately 1,000 import records for these products but less than two dozen independently-negotiated tariff rates. Thus, the importance of these negotiated changes would be overstated if we include every product with a Schedule A number. Since the tariff reduction of these products is systematic, the tariff reductions on these products can be summarized by a few representative lines. Accordingly, in our analysis, we have included only the Schedule A code for the product

with the median import value within each subgroup from 3010 to 3067 (excluding 3030). In this manner, the changes in tariff rates for the textile group products are captured, and the risk of overemphasis is limited. To ensure the reliability of our findings, we conducted a robustness check by separately testing the inclusion of all textile products. Our primary result remained consistent even after this additional analysis.

3.3 Handkerchiefs

A large number of cotton handkerchief products have their tariff rate set as the base rate of the cotton cloth that is their chief component. Thus, again, changes in their tariff rates are entirely dependent on the negotiations on cotton cloth. As all products utilized in the production of these handkerchiefs have already been included in our representative lines for cotton cloth, entering separate lines for each one for the handkerchief versions would be redundant. Moreover, as with the textile products, there are a considerable number of import records with minimal import values for these dependent handkerchief products. Accordingly, we only retain the handkerchief products that have explicitly-specified, independently-negotiated tariffs.

3.4 Mixed Tariffs

We classify a product as having a mixed tariff when that product has a minimum or maximum tariff in addition to the usual ad valorem or specific tariff. For example, Schedule A code 5380550 in the 1950 Schedule A (Paragraph Number 202a) *Earthen tiles: Quarries or quarry tiles: Valued not over 40 cents per square foot and more than 5/8 inch in thickness* has a

Smoot-Hawley rate of 10 cents per square foot, and the minimum ad valorem rate of 50 percent and the maximum ad valorem of 70 percent. In the bilateral agreement with the United Kingdom, the rates for this product were reduced to 5 cents per square foot, the minimum ad valorem rate of 25 percent, and the maximum ad valorem rate of 70 percent. After the Geneva round, the tariff rates for this line were further reduced to 5 cents per square foot, the minimum ad valorem rate of 15 percent, and the maximum ad valorem rate of 70 percent. When merging this tariff schedule with the import data, we take account of these cases by using the AVEs for any specific portion of the tariff specification to determine the applied tariff and only enter the applied tariff in the final data.

3.5 Line Splitting

Line splitting is usually associated with the introduction of new product distinctions. Typically, new lines are created when negotiators agree to heterogeneous tariff reductions within what was previously one Schedule A code. For example, the product *“Prepared or preserved: Birds, including poultry, prepared or preserved, n.s.p.f.”* was assigned the Schedule A code 0026000 in the 1946 Schedule A with a rate of 10 cents per pound. Because of the Geneva round negotiations, the 1950 Schedule A saw a division into two distinct lines: (i) *“Turkeys, prepared or preserved,”* with code 0026500 and rate of 5 cents per pound; (ii) *“Prepared or preserved: Birds, including poultry, prepared or preserved, n.s.p.f.”* with code 0026900 and a rate of 10 cents per pound. In this case, the splitting occurs because a more specific product (turkey) within a broader category (poultry) was negotiated separately while other products under this category remained unchanged. To deal with these kinds of changes to the Sched-

ule A, we have two lines in our data with identical 1946 Schedule A codes, Smoot-Hawley tariff rates, and pre-Geneva tariff rates, but different tariff rates for the Geneva round.

We must take account of Schedule A code changes when calculating the AVEs for a given year. This is because we use previous-year import data in calculating AVEs to avoid having the tariff changes impact the unit values. Since we cannot separate the quantities and values in the previous-year import data for lines that split, we apply the unit value for the previous, un-split Schedule A number to each new Schedule A code that was split out from the original.

4 Bilateral Trade Agreements

4.1 Country-by-Country Detail

Between 1934 and 1946, the U.S. concluded a series of bilateral trade agreements that became the model for the GATT negotiations. The distribution of these negotiations by country can be seen in Table 5.

4.2 Trade Agreement with Paraguay

The U.S. and Paraguay signed a bilateral trade agreement on September 12, 1946. The agreement was proclaimed by President Truman on March 10, 1947 and became effective on April 9, 1947. The negotiations, therefore, did not overlap with those of the GATT in Geneva that began on April 10, 1947.

The U.S. list of concessions covers 22 paragraphs; these map onto 49 Schedule A numbers.

Table 5: U.S. Bilateral Trade Agreements

Country	Date Effective	Number of Products	GATT?
United Kingdom	Jan. 1, 1939	909	Yes
Canada	Jan. 1, 1936, Jun. 17, 1939	398	Yes
Mexico	Jan. 30, 1943	239	
Belgium	May 1, 1935	197	Yes
Switzerland	Feb. 15, 1936	182	
France	June 15, 1936	179	Yes
Argentina	Jan. 8, 1943	151	
Peru	July 29, 1942	138	
Sweden	Aug. 5, 1935	116	
Netherlands	Feb. 1, 1936	99	Yes
Uruguay	Jan. 1, 1943	55	
Iran	Jun. 28, 1944	51	
Finland	Nov. 2, 1936	37	
Brazil	Jan. 1, 1936	31	Yes
Turkey	May 5, 1939	26	
Iceland	Nov. 19, 1943	23	
Costa Rica	Aug. 2, 1937	22	
Venezuela	Dec. 16, 1939	19	
Guatemala	June 15, 1936	18	
Ecuador	Oct. 23, 1938	17	
Haiti	June 3, 1935	14	
Colombia	May 20, 1936	14	
Honduras	Mar. 2, 1936	12	
El Salvador	May 31, 1937	8	

Notes: Effective dates are from the texts of each bilateral agreement. Number of products are counts from the 1946 Schedule A as verified in the bilateral agreement texts. The GATT column denotes contracting parties to the original GATT in 1947. A trade agreement with Nicaragua was enacted on October 1, 1936 and terminated on March 10, 1938. A trade agreement with Czechoslovakia was enacted on April 15, 1938 and terminated on April 22, 1939. A trade agreement with Paraguay was enacted on April 9, 1947 but only changed the tariff on one product (urunday extract). We exclude the agreements with Nicaragua, Czechoslovakia, and Paraguay from our analysis. We also exclude the agreement with Cuba since commitments to Cuba were not extended as MFN tariff commitments.

Almost all of these concessions had previously been made to other bilateral trading partners.

The trade agreement bound at zero the tariffs of four products that were free under Smoot Hawley (bark of curupay, pettigrain, hides of wild pig and hog, and undressed furs of jaguar).

One product, urunday extract, had its tariff changed. If it was classifiable under Paragraph

38, the tariff was reduced from 15% to 7.5%. If it was not classifiable under Paragraph 38, the tariff was reduced to 10%.

We did not create separate variables for 1947 to take account of these two changes. Instead, we simply incorporated these two tariff changes in our Geneva Round data. The change to 7.5% is accounted for under Schedule A number 2345 710 and the change to 10% is accounted for under Schedule A number 2950 030.

4.3 Import data

We need product-level U.S. import data for two purposes: first, the import value for each product is crucial for measuring the importer market power; second, specific tariffs are quite common in this era and we use import values to calculate AVEs for products with specific duties. Following the previous example of tariff rates on *Rye*, the import entry for *Rye* in year 1946 is shown in Figure 8.

VOLUME I—FOREIGN COMMERCE AND NAVIGATION

83

NO. 1—IMPORTS FOR CONSUMPTION OF MERCHANDISE BY COMMODITY BY COUNTRY OF ORIGIN, 1946

Commodity Group 1—Vegetable Food Products and Beverages

1020000-1090260

BARLEY (BUSHEL 48 POUNDS)			RICE MEAL, FLOUR, POLISH, AND BRAN			WAFERS, PUDDINGS, CAKES AND OTHER BAKED ARTICLES (INCLUDING SHEETED OR FLAVORED BISCUITS)			PEARL BARLEY		
Tar. Par. 722 15¢ bu. A.R.			Tar. Par. 727 5/8¢ lb.			Tar. Par. 733 15¢ A.R.			Tar. Par. 722 1¢ lb. A.R.		
Cal. Duty \$661,709 Eq. Ad Val. 10.2%			Cal. Duty \$23,752 Eq. Ad Val. 19.4%			Cal. Duty \$1,252			Cal. Duty \$4 Eq. Ad Val. 1.8%		
A No. 1020 000			A No. 1059 100			A No. 1075 500			A No. 1090 150		
Bushels Dollars			Pounds Dollars			Dollars			Pounds Dollars		
Total..... 4,411,433 6,499,136			Total..... 3,800,372 122,369			Total..... 8,345			Total..... 394 221		
Canada..... 36 142			Mexico..... 1,914,800 22,067			Canada..... 1,995			Canada..... 100 3		
Canada..... 4,411,395 6,498,996			Dominican Rep..... 100 15			Newfoundland Labr..... 153			China..... 294 213		
			Colombia..... 3,157 193			Norway..... 2,014					
			Brazil..... 1,873,360 95,835			Denmark inc Faroe..... 65					
			Iran..... 441 100			United Kingdom..... 2,515					
			China..... 8,542 3,668			Netherlands..... 736					
			Hong Kong..... 572 271			France..... 1					
						Portugal..... 1					
						China..... 349					
						Hong Kong..... 516					

Figure 8: Import value and quantity data for Rye, 1946